# TRAN NHAT NAM

*Product Data Engineer | Ho Chi Minh City, Vietnam*

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## Summary

**Product Data Engineer** with 4+ years of experience creating **end-to-end data solutions** – from **ETL/ELT pipelines** and **data platforms** to **data governance, BI and AI applications**. More than a technician: I dive deep into the business needs of each department (Sales, Marketing, Finance, BOD) to turn data into decisions – **cutting ETL operational cost by 90%**, **consultation time by 60%**, and **accelerating decisions up to 3x**.

## Technical Skills

- **Languages:** Python, SQL
- **Data Engineering:** dbt (medallion modeling, tests, docs), Dagster, Airflow, dlt, Airbyte, n8n, Apache Kafka, ETL/ELT
- **Databases & Warehouses:** PostgreSQL, BigQuery, DuckDB, StarRocks, dimensional modeling (star schema), data lineage, vector databases
- **Lakehouse & Processing:** Apache Iceberg, Apache Spark (PySpark), Trino, Polaris, MinIO, Parquet
- **BI & AI/ML:** Looker, Metabase, LLM APIs, RAG & vector search, prompt engineering, HuggingFace, ASR (Wav2Vec 2.0)
- **Cloud, DevOps & Deployment:** AWS Lambda, Google Kubernetes Engine (GKE), Helm, Docker, GitHub Actions, CI/CD, Linux, Vercel, Railway, Dokploy, Coolify
- **Integration & Business Systems:** REST APIs, web scraping (Scrapy, BeautifulSoup), Odoo, ERPNext, Haravan, BizFly, AppSheet
- **Stakeholder Communication:** translating business requirements into technical specs, executive presentations, self-serve dashboards

## Experience

**Product Data Engineer | Jemmia Diamond** **Sep 2023 – Present**

- **[Data Platform & Orchestration]** Architected and deployed a Modern Data Stack on Dagster (via Dokploy) as software-defined assets, with dlt/Airbyte/n8n ingestion centralizing 12+ sources (Social Ads, Haravan, ERP, Stringee) into a PostgreSQL Data Warehouse – 150 raw tables processing 500K+ rows/day, containerised with Docker Compose.
- **[Data Modeling & Transformation]** Designed Star Schema (Kimball) models and authored 200+ dbt transformations across Raw → Staging → Intermediate → Marts, cutting development and maintenance time by 90% with dbt tests (schema, not-null, relationships, freshness), PII masking and lineage via OpenMetadata.
- **[Serverless ETL & IoT Integration]** Migrated brittle legacy workflows (Zapier) to event-driven architectures on AWS Lambda (deployed via AWS SAM). Designed a high-throughput pipeline integrating hardware RFID scanners with e-commerce systems via webhooks (~1,000 records / 3MB per scan, 20 scans/day), enabling time-series analysis for inventory optimization and supply-chain management.
- **[Enterprise Systems & Master Data]** Led enterprise systems transformation: migrated 10GB+ to ERPNext and engineered a custom PIM on NocoDB (40+ tables) over a Cloudflare R2 data lake, with zero-downtime backfill for 8,000+ SKUs and CDC – cutting cross-departmental manual operations 40% at 97% adoption.
- **[Business Intelligence & Analytics]** Acted as the primary analytics partner for Sales, Marketing, Finance and the BOD – gathering business requirements, defining KPIs and a governed semantic layer, and translating them into 30+ executive dashboards in Looker (100% coverage of key metrics). Migrated the BI stack to Metabase to deliver self-service analytics with standardized metric definitions, accelerating data-driven decisions up to 3x and cutting BI costs by retiring Looker licensing.
- **[GenAI & Data Activation]** Built AI products on the warehouse: **Salesaya** (AI sales assistant cutting consultation time 60%) and **Metabot** – a Telegram/Discord agent for secure natural-language data queries without exposing the underlying database (Metabase API, Next.js, React, Redis context caching).
- **[Storage & Disaster Recovery]** Architected a highly available NAS with automated off-site replication to Google Drive for disaster recovery, and provisioned OpenVPN tunnels with strict RBAC for secure remote access.
- **[DataOps & Leadership]** Established CI/CD pipelines for all analytical applications and ERP deployments, and configured real-time anomaly alerting via Sentry. Led data-literacy workshops for the BOD and department heads, driving a data-driven culture and engineering standards across the organization.

**Data Analyst (Contract) | BabyBet (Startup)** **Jul 2023 – Sep 2023**

- **Contracted to assess a mobility venture's feasibility:** consolidated market & competitor data on 5+ players (Grab, be) from 15+ market-intelligence, survey and secondary-research sources into structured datasets.
- **Delivered 10+ demand, willingness-to-pay, pricing and unit-economics analyses** that informed the cross-functional go/no-go decision and phased strategy.

**Data Analyst | AVLI Group JSC** **Feb 2023 – Jul 2023**

- **Built the ingestion layer:** AWS Lambda scraping workers (Python, BeautifulSoup, REST APIs) for batdongsan.com and 7+ other sites, plus the company's CSV/Excel/Sheets file sources via **Airbyte** – loading 300K+ property records into PostgreSQL.
- **Designed a 15-table dimensional data warehouse** (star schema) as a single source of truth for customers, properties and interactions, orchestrated end-to-end with **Airflow**.
- **Built a low-code CRM on AppSheet** (customer tracking, attendance, call logging for customers/landlords, apartment images, care history) – **cutting customer-care and property-lookup time by 50%**.
- **Delivered 10+ Looker dashboards** for customer demand, inventory and sales performance, used by 25+ users.

**Data Research Assistant | AISIA Research Lab** **Jun 2022 – Jan 2023**

- Labeled 80+ hours of Vietnamese speech data and helped optimize **ASR models (Wav2Vec 2.0)**, cutting word error rate by ~**12%** to narrow the gap with FPT's solution.

## Personal Projects

**Mini Lakehouse (Apache Iceberg Lab)** **2026**
*Technologies: Apache Iceberg, Spark (PySpark), Trino, Polaris, MinIO, Parquet, Docker* repo

- Built a multi-engine lakehouse where Spark and Trino share Apache Iceberg tables via a Polaris REST catalog over MinIO, covering partitioning, time-travel, COW/MOR and table maintenance end-to-end.

**FB Feed Crawler (Multi-Provider Scraping Platform)** **2026**
*Technologies: Python, FastAPI, PostgreSQL, APScheduler, Docker, Playwright/Crawl4AI* repo

- Built a resilient Facebook crawler with a 3-engine fallback chain (Crawl4AI, Scrapling, facebook-scraper) behind circuit breakers, plus daily scheduling, crash recovery and a FastAPI management API.

**CryptoStream Labs (Automated ELT Pipeline)** **2024**
*Technologies: GCP (GKE, BigQuery), Apache Airflow, dbt, Airbyte, Docker, Helm, SQL* repo

- Built an ELT pipeline ingesting crypto data via Airbyte into BigQuery, transformed with dbt (partitioned OBT), orchestrated on Airflow and deployed on GKE via Helm.

**Hackathon & Open-Source Side Projects** **2024–2026**
*Technologies: Next.js, TypeScript, Docker, Airflow, Airbyte, pg_duckdb, StarRocks* repos

- **Hackathon:** built SAIGONXUA, an anti-marketing jewelry app (Next.js, Gemini) that tells shoppers why to buy or avoid each product.
- **Open source:** Docker templates running data tools on lightweight PaaS/VPS – Airflow, Airbyte, pg_duckdb, StarRocks.

**Sign2Text (Traffic-Sign Recognition – University Research)** **2022**
*Technologies: Python, YOLOv7, VietOCR, OpenCV, Streamlit* repo

- Led a university scientific-research team building a traffic-sign recognition system – labeled the dataset, trained a YOLOv7 detector, and extracted Vietnamese text from signs with VietOCR.

## Education

**University of Information Technology, VNU-HCM** **2019 – Jun 2023**
Bachelor of Data Science Ho Chi Minh City, VN
GPA: 7.9/10 | TOEIC 725